

# **Aging impairs the temporal clustering of sleep spindles.**

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During the second half of our lives and starting already within our forties, our sleeping patterns change. These changes might initially not appear as dramatic as the reorganization of sleep during infancy and early childhood, but they can greatly affect our subjective quality of life. On the one hand, these alterations occur on the macro scale of overall sleep architecture and might entail a general fragmentation of sleep, a higher number of arousals and reduced amounts of slow wave sleep. On the other hand, age impacts sleep on the microscale such that the overall frequency composition, cross frequency coupling as well as the occurrence of sleep oscillations in the EEG signal is affected. For instance, it has been shown that with aging slow oscillations display reduced amplitudes, a flatter slope and lower densities [1,2] and that sleep spindles are reduced in number and duration [3].

In this issue of SLEEP Champetier et al.[4] investigated how aging affects the temporal distribution of sleep spindle occurrence and how this relates to memory consolidation. Based on a method by Boutin and Doyon [5], they considered spindles that were temporally clustered within the same EEG channel at intervals shorter than 6 seconds as spindle trains. Importantly, Champetier et al.[4] focused their exploratory EEG analysis on fast spindles (13-16 Hz) that had been detected using participant-individual peaks in the power spectrum of the centro-parietal electrodes. They included EEG data from 134 elderly participants (65-83 years old) and 18 young to middle aged adults (20-52 years) while memory performance was assessed only in elderly participants (n = 57) using a visuospatial object location task. Their results show that aging is associated with a reduced rate of clustered spindles that drops from around 75% in 20 years old to 60 % in 80 years old participants. This effect was particularly driven by the fact that elderly people displayed substantially less spindle trains with four or more consecutive spindles. Additionally, the fewer spindles occurred within one train, the lower was their mean amplitude and duration, but the higher their oscillatory frequency. Interestingly, these dynamics did not differ substantially across middle-aged and elderly participants.

It is well established that sleep spindles are associated with memory consolidation during sleep [6]. Ample evidence has shown that local spindle density is higher over brain regions displaying higher activity levels during preceding awake memory encoding [7] and that particularly sleep spindles that occur during the depolarizing up-state of slow oscillations serve memory consolidation [6]. However in the elderly, Champetier et al.[4] did not find a general association between overall centro-parietal spindle density and memory recall after sleep, but instead identified a significant correlation between the mean number of consecutive spindles per train and overnight memory performance. This was not the case for the proportion of temporally clustered spindles relative to all spindles. Therefore, it appears that the beneficial effect of spindle trains on memory consolidation is even greater than that of the same number of spindles occurring solitarily.

The findings by Champetier et al. [4] raise the question what the underlying neuronal mechanisms for this beneficial effect of spindle train length on memory consolidation might be. First, sleep spindles generally provide a time window that promotes synaptic plasticity by facilitating dendritic calcium influx [8] and have been proposed as a mechanism facilitating hippocampo-neocortical communication by coordinating hippocampal and cortical memory

reactivation [6]. However, the pattern of cells that are active during sleep spindles changes with time [9]. Therefore, spindles occurring in shorter intervals might reactivate pattern that are more similar, thereby improve the signal to noise ratio of reactivated memories, compared to spindles that occur more distant in time. At the same time sleep spindles typically alternate with refractory periods, which have been proposed to be important for reprocessing of memory reactivation occurring during the preceding spindles by protecting these reactivated memories from interference [10]. Potentially with aging solitary spindles and their following refractory periods alone are not sufficient to serve this function anymore. Additionally, the fact that the current study did not find correlations between spindle density and memory consolidation might be owed to the focus on central electrodes, which might not have captured the reactivation processes occurring in task-relevant regions.

A second underlying mechanism might rely on the phase amplitude coupling of spindles and slow oscillations. Previous experiments could show that their coupling changes with aging, correlates with brain degeneration in medial prefrontal cortex and is predictive for memory performance after sleep [11,12]. Additionally, experiments in mice revealed that those spindles nested in slow oscillation up-states are accompanied by a unique constellation of cortical excitation and inhibition. Here the released dendritic inhibition of cells that are reactivated during spindles occurring in the slow oscillation up-state might particularly promote synaptic plasticity on apical dendrites. [13]. It's currently an open question how the temporal clustering of sleep spindles relates to their co-occurrence with slow oscillations..

Third, with aging the locus coeruleus (LC), the sole source of noradrenaline (NA) in our brain often shows signs of neurodegeneration and reduced numbers of noradrenergic cells [14]. Recent animal work has shown that NA activity fluctuates in an infraslow rhythm of  $\sim 0.02$  Hz during non-rapid eye movement (REM) sleep, partitioning sleep in phases with high and low spindle activity [15,16]. Disturbing this fluctuation impairs memory consolidation [15,17] and can cause transitions into wakefulness or REM sleep [15,16]. Champetier et al. [4] detected comparable infraslow modulations of spindle activity in middle aged and elderly participants. However, they also found that slower infraslow fluctuations were associated with a higher number of consecutive spindles within one train as well as with a higher overall ratio of temporally clustered spindles. This finding supports the idea that NA fluctuations might also affect the temporal occurrence of sleep spindle trains, which is impaired in the elderly.

Interestingly, when splitting the night in half, the association between the number of consecutive spindles within spindle trains and memory consolidation was only significant for the second half of the night [4], during which more Non-REM to REM transitions occur [6,18]. The fact that REM sleep always follows Non-REM sleep might result from the necessity that sleep spindles selectively upregulate cellular activity that is important for memory consolidation while REM sleep afterwards more generally downregulates cortical activity [9]. This downregulation would – without the previous spindle activity- lead to a disruption of memory traces. Such disrupting effects of REM sleep on memory could occur in elderly as they spent similar time in REM sleep compared to young adults while the

amount of slow wave sleep is reduced [18] and thereby the protective function of sleep spindles for memory might be less pronounced.

With their findings of age-related and memory-relevant changes in spindle train length, Champetier et al.<sup>4</sup> inspire future experiments that might unravel the exact mechanisms that underlie the temporal clustering of spindles and how the reoccurrence of spindles as spindle trains particularly fosters memory consolidation during sleep.

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## Disclosure Statement

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